

Problem D. Maximum modulo equality

Time Limit 5000 ms

Mem Limit 262144 kB

You are given an array a of length n and q queries l, r .

For each query, find the maximum possible m , such that all elements $a_l, a_{l+1}, \dots, a_r$ are equal modulo m . In other words, $a_l \bmod m = a_{l+1} \bmod m = \dots = a_r \bmod m$, where $a \bmod b$ — is the remainder of division a by b . In particular, when m can be infinite, print 0.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two integers n, q ($1 \leq n, q \leq 2 \cdot 10^5$) — the length of the array and the number of queries.

The second line of each test case contains n integers a_i ($1 \leq a_i \leq 10^9$) — the elements of the array.

In the following q lines of each test case, two integers l, r are provided ($1 \leq l \leq r \leq n$) — the range of the query.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$, and the sum of q does not exceed $2 \cdot 10^5$.

Output

For each query, output the maximum value m described in the statement.

Examples

Input	Output
3 5 5 5 14 2 6 3 4 5 1 4 2 4 3 5 1 1 1 1 7 1 1 3 2 1 7 8 2 3 1 2	3 1 4 1 0 0 1 6

Note

In the first query of the first sample, $6 \bmod 3 = 3 \bmod 3 = 0$. It can be shown that for greater m , the required condition will not be fulfilled.

In the third query of the first sample, $14 \bmod 4 = 2 \bmod 4 = 6 \bmod 4 = 2$. It can be shown that for greater m , the required condition will not be fulfilled.